

**Table 61. IWDG min/max timeout period at 32 kHz (LSI)**

For the values in this table, the exact timings still depend on the phasing of the APB interface clock versus the LSI clock, so that there is always a full RC period of uncertainty.

| Prescaler divider | PR[2:0] bits | Min timeout RL[11:0] = 0x000 | Max timeout RL[11:0] = 0xFFFF | Unit |
|-------------------|--------------|------------------------------|-------------------------------|------|
| /4                | 0            | 0.125                        | 512                           | ms   |
| /8                | 1            | 0.250                        | 1024                          |      |
| /16               | 2            | 0.500                        | 2048                          |      |
| /32               | 3            | 1.0                          | 4096                          |      |
| /64               | 4            | 2.0                          | 8192                          |      |
| /128              | 5            | 4.0                          | 16384                         |      |
| /256              | 6 or 7       | 8.0                          | 32768                         |      |

**Table 62. WWDG min/max timeout value at 144 MHz (PCLK)**

| Prescaler | WDGTB | Min timeout values | Max timeout value | Unit |
|-----------|-------|--------------------|-------------------|------|
| 1         | 0     | 0.0284             | 1.820             | ms   |
| 2         | 1     | 0.0569             | 3.641             |      |
| 4         | 2     | 0.1138             | 7.282             |      |
| 8         | 3     | 0.2276             | 14.564            |      |
| 16        | 4     | 0.4551             | 29.127            |      |
| 32        | 5     | 0.9102             | 58.254            |      |
| 64        | 6     | 1.820              | 116.508           |      |
| 128       | 7     | 3.641              | 233.017           |      |

### 5.3.22 I3C interface characteristics

The I3C interface meets the timing requirements of the MIPI® I3C specification v1.1.

The I3C peripheral supports:

- I3C SDR-only as controller
- I3C SDR-only as target
- I3C SCL bus clock frequency up to 12.5 MHz

The parameters given in Table 63 below are obtained with the following configuration:

- Output speed is set to OSPEEDRy[1:0] = 10
- I/O Compensation cell activated
- Voltage scaling range 1

The I3C timings are in line with the MIPI specification, except for the ones given in Table 63, I3C open-drain measured timing. For  $t_{SU\_OD}$ , this can be mitigated by increasing the corresponding SCL low duration in the I3C\_TIMINGR0 register. For further details refer to AN5879.

**Table 63. Open drain timing measurements**

Evaluated by characterization - Not tested in production.

| Symbol       | Parameter                              | Conditions                                                    | Min               | Unit |
|--------------|----------------------------------------|---------------------------------------------------------------|-------------------|------|
| $t_{SU\_OD}$ | SDA data setup time in open drain mode | Controller<br>$2.7\text{ V} \leq V_{DDIOX} \leq 3.6\text{ V}$ | 22 <sup>(1)</sup> | ns   |

1. The minimum SDA data setup time during open-drain-mode is 3 ns, as specified in the MIPI Alliance specification for I3C.